# Pricing of a CAT bond

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# 1 ACE as an Index Underlying a Catastrophe Bond

The chosen underlying index is the Atlantic Hurricane Accumulated Cyclone Energy (ACE) index. ACE is an index used to measure the activity of a cyclone/hurricane season. It combines the number of hurricane systems, how long they existed and how intense they became. This is a real-life and non-tradable catastrophe-related index that measures the cumulative energy generated by tropical storms and hurricanes in the North Atlantic Basin. It is as a time series of ACE for the Atlantic Ocean, with data available from 1851 to 2022, which provides a long historical sample suitable for analysis [1]. The data was downloaded from Our World in Data (https://ourworldindata.org/grapher/accumulated-cyclone-energy). The ACE index is calculated by multiplying a tropical cyclone's maximum sustained winds every six hours to generate the values. The total sum of these values is calculated to get the total for a storm and is divided by 10000 to make it more manageable. The ACE index includes all tropical cyclones with winds above 35 knots (65 km/h). This ACE index is the sum of all tropical cyclones and hurricanes during a year in the North Atlantic Basin area.

The ACE index has no or only low correlation with financial markets due to different underlying drivers. The ACE index is driven by atmospheric physics and ocean temperatures, while the financial markets are driven by economic expectations. The ACE index's only possible correlation with financial markets would be from a high impact hurricane that changes investors' economic expectations. However, the aggregate macroeconomic impact is only limited because the damage is local. Corresponding CAT bonds would therefore also be uncorrelated with the market returns, so-called "zero-beta assets". This correlation is important because investors seek to diversify their portfolios, and CAT bonds based on uncorrelated indexes like the ACE index make this diversification possible.

The reverse relationship is equally limited. Even in the most extreme ACE years, financial markets cannot influence the ACE index, as interest rates, company earnings, or investor behavior cannot alter sea surface temperatures or atmospheric conditions. Furthermore, even the most destructive hurricane seasons do not produce lasting broad market shocks. The most active Atlantic hurricane season ever recorded was 2005, with Hurricanes Katrina, Rita, and Wilma all making landfall and causing over $ 160 billion in damage [2, 3], yet the SP 500 still ended 2005 with a positive annual return [4]. Any market impact is therefore sector-specific, concentrated in insurers and reinsurers, rather than a systematic market-wide effect, confirming the zero-beta nature of ACE-linked CAT bonds.

The ACE index is specifically associated with tropical cyclone risk in the North Atlantic Basin. The catastrophic risks associated with tropical cyclones include physical destruction risk with wind damage to buildings, infrastructure destruction and agricultural losses and flooding and storm surge risk with coastal flooding and inland flooding from heavy rainfall. These catastrophic risks have a high economic impact with insurance claims and business interruption. Tropical cyclones are among the most destructive natural disasters worldwide (AMETSOC). According to the World Bank, a quarter of global GDP is exposed to substantial economic losses from these events (Dilley et al., 2005).

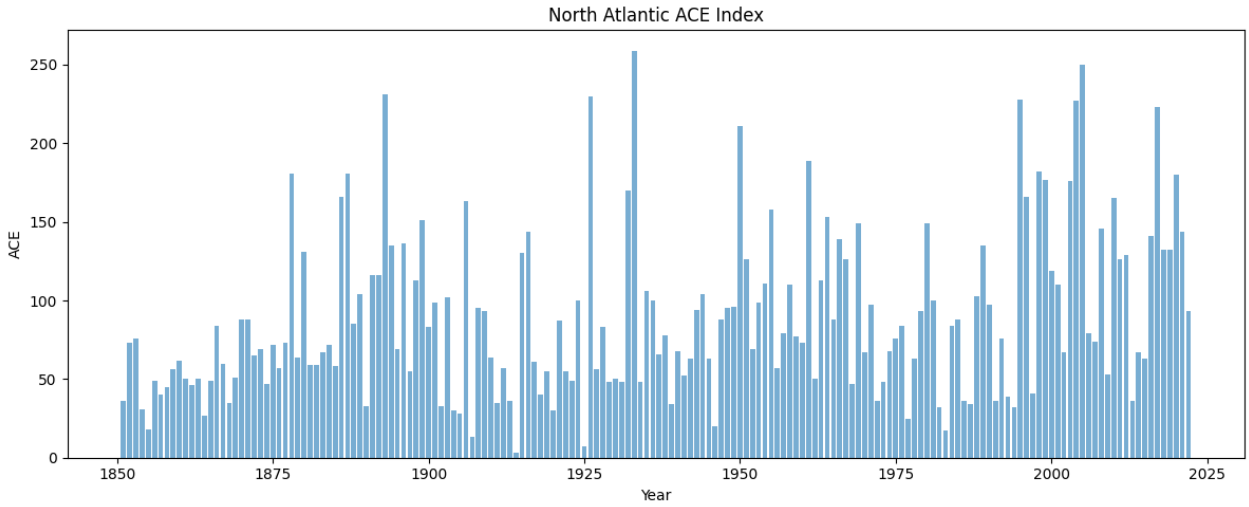



Figure 1: North Atlantic ACE Index, 1851-2022

The ACE index exhibits substantial year-to-year volatility, with values ranging from approximately 20 to over 250. However, beyond this short-term volatility, the ACE index displays a clear long-term periodicity driven by the Atlantic Multi-decadal Oscillation (AMO): a natural cycle of warming and cooling in North Atlantic sea surface temperatures with a period of approximately 60 years [5]. During warm AMO phases, sea surface temperatures are higher, providing more energy to developing storms and producing more active hurricane seasons. During cool AMO phases, activity is lower [6]. This periodicity is visible in the data: the periods roughly spanning 1926-1970 and 1995-present correspond to active (warm AMO) phases, while the period from approximately 1971-1994 stands out as a quiet phase, with consistently lower ACE values across nearly the entire stretch [6].

Several such periods can be observed. The 1933 season produced one of the highest ACE values of the early 20th century, corresponding to the active 1933 Atlantic hurricane season which saw 21 tropical storms, making it the most active season recorded up to that point [7]. The 1930s stand out due to a pronounced peak, reaching around 260, which can be considered an extreme outlier. The post-1995 period contains the most prominent spikes in the data: the 2005 season produced an exceptionally high ACE value, directly attributable to Hurricane Katrina, Rita, and Wilma all occurring within the same season, 2005 remains the most active Atlantic hurricane season ever recorded [3]. The 2004 and 2010 seasons also show high values, consistent with the warm AMO environment [8].

In this recent period, high ACE values occur more frequently, and extreme years are more common compared to earlier decades. This increase in activity is primarily explained by the return to a warm AMO phase in 1995 [6]. Additionally, climate change may be contributing to greater storm intensity, as warmer global sea surface temperatures provide more energy for cyclone development, even if the total number of storms does not necessarily increase [9].

Before turning to the risk-management application, it is important to note that CAT bonds also provide a direct link between physical climate risk and financial pricing. If the frequency or severity of extreme weather changes over time, historical catastrophe distributions may become less rep-

resentative of future risk. This creates model uncertainty for both issuers and investors and makes CAT bonds a relevant instrument for studying how climate-related physical risk is transferred and priced in capital markets.

## 2 ACE CAT Bond from a Risk Management Perspective

A catastrophe (CAT) bond written on the ACE index can be a valuable instrument for risk management, as it allows insurers to transfer exposure to extreme tropical cyclone activity to capital markets. In such structures, investors purchase bonds issued by insurers and receive a premium for bearing catastrophe risk [10].

If a predefined trigger is activated, when a catastrophe event occurs that meets specific criteria regarding the ACE index, the insurer is only required to repay a fraction of the bond's principal to investors. The remaining portion of the principal can then be used to cover insurance claims arising from the disaster. In this way, cat bonds provide insurers with an additional source of capital precisely when losses materialize.

CAT bonds on the ACE index are useful because of the nature of catastrophic risk. Tropical cyclones generate low-frequency but high-severity losses that are difficult to diversify within traditional insurance portfolios [11]. By issuing a CAT bond, insurers can offload part of this tail risk to external investors. Moreover, because CAT bonds are typically uncorrelated with financial markets, they provide investors with diversification benefits, making it easier for insurers to attract capital.

A CAT bond is particularly useful during periods of increased catastrophe risk or when traditional reinsurance becomes expensive or capacity-constrained. After major disaster events, reinsurance premiums often increase significantly, making alternative risk transfer mechanisms more attractive [12]. In such environments, CAT bonds provide a stable and potentially more cost-effective source of protection. In addition, they are especially relevant when insurers are exposed to systemic climate-related risks, such as increasingly intense hurricane seasons. The actual value of CAT bonds materializes when a catastrophic event occurs and the trigger is activated.

How a CAT bond based on the ACE index functions depends on its trigger design. Several types of triggers are commonly used in practice. Indemnity triggers are based on the actual losses incurred by the issuing insurer, making them closely aligned with traditional reinsurance [10]. Industry loss triggers rely on estimates of total losses across the insurance industry, typically provided by an independent third-party modeler. Parametric triggers are based on the physical characteristics of the event, such as hurricane wind speed, barometric pressure, or earthquake magnitude, allowing for rapid and transparent payouts [13]. In the case of the ACE index, this is especially advantageous as it is calculated and published independently by NOAA, making it verifiable and resistant to manipulation, thereby reducing moral hazard relative to indemnity-based structures [10]

In this case, the bond would operate as a parametric instrument, where payouts are triggered when the ACE index exceeds a predefined threshold. Since the ACE index measures the cumulative intensity and duration of tropical cyclones, it offers an objective and transparent indicator of overall storm activity. This allows for rapid payouts without the need for lengthy loss assessments, which is particularly valuable in the aftermath of catastrophic events when liquidity is crucial.

However, the effectiveness of such a structure depends critically on the calibration of the trigger. A key limitation of parametric CAT bonds is basis risk, the possibility that the ACE index exceeds the threshold without causing significant losses to the insurer, or that severe losses occur without triggering the bond [10]. Therefore, careful design of the threshold and contract parameters is essential to ensure that payouts align with the actual losses.

A CAT bond written on the ACE index can enhance risk management by providing additional capital, improving risk transfer, and ensuring timely liquidity following extreme events. Its effectiveness, however, depends on appropriate trigger design and the extent to which the index accurately captures the insurer's underlying exposure.

# 3 ACE Index Following a Poisson Jump-Diffusion

To estimate the parameters of the Poisson Jump-Diffusion (PJD) model, we used the code in `part2_cat_est.R`. The following changes were made to adapt it to our ACE data:

- The data is loaded from a CSV file downloaded from Our World in Data, covering the period 1851 to 2022, giving 171 annual observations and 170 log returns.
- Since the ACE index consists of strictly positive values, no transformation was needed before taking logarithms.
- The time step was changed from monthly to annual by setting delta.t = 1, as the ACE index is recorded once per year.
- A sensitivity analysis was done on epsilon by testing values of 0.8, 1.0, and 1.2 to examine how the classification of jumps versus diffusion observations changes. Based on this analysis, described below, we chose $\epsilon = 1.0$.

The following table summarises the sensitivity analysis on epsilon:

| $\epsilon$ | Number of jumps | Number of diffusion obs. | $\hat{\lambda}$ |
|---|---|---|---|
| 0.8 | 49 | 122 | 0.287 |
| 1.0 | 35 | 136 | 0.205 |
| 1.2 | 20 | 151 | 0.117 |

**Table 1:** Sensitivity analysis $\epsilon$

For $\epsilon = 0.8$, nearly 29% of all years are classified as jump years, which is too frequent for events that should represent truly exceptional hurricane seasons. For $\epsilon = 1.2$, only 20 jump observations remain, which is quite small for reliable estimation of the jump parameters [14]. We therefore chose $\epsilon = 1.0$ as it strikes a balance between having enough jump observations (35) for reliable estimation and keeping the jump intensity at a sensible level of approximately one jump every five years.

The following table reports the MLE estimation results:

| Parameter | Estimate | Std. Error | $t$-value | $p$-value |
|---|---|---|---|---|
| $\mu$ | 0.196 | 0.094 | 2.093 | 0.036 |
| $\sigma$ | 0.660 | 0.076 | 8.722 | $< 0.001$ |
| $\delta$ | 1.550 | 0.884 | 1.752 | 0.080 |
| $\lambda$ | 0.135 | 0.142 | 0.954 | 0.340 |
| $\kappa$ | 0.202 | 0.452 | 0.447 | 0.655 |

**Table 2:** Maximum Likelihood Estimation Results for PJD Parameters

The log-likelihood at the optimum is -208.94.

All five estimated parameters have signs and magnitudes that are economically meaningful.

$\mu = 0.196$ is positive and statistically significant at the 5% level. This indicates a slight upward drift in the ACE index over time, which is consistent with the return to a warm AMO phase from 1995 onwards and the potential influence of climate change on storm intensity [9]. It should however be noted that $\mu$ is generally the hardest parameter to estimate accurately, and its exact value should be interpreted with caution.

$\sigma = 0.660$ is positive and highly significant, with the smallest standard error of all parameters. This represents the typical year-to-year volatility of the ACE index in the absence of jumps. It is consistent with the observed distribution of log returns, where the middle 50% of observations ranged from approximately -0.52 to +0.42.

$\delta = 1.550$ is the standard deviation of jump sizes and is marginally significant at the 10% level. Note, $\delta > \sigma$, meaning that when jumps do occur, they are more variable than ordinary diffusion moves [15]. This makes intuitive sense for the ACE index: extreme hurricane seasons vary enormously in severity, from moderately active years that just exceed the threshold to historically exceptional seasons such as 2005 with Hurricane Katrina, Rita, and Wilma all occurring within the same season [3].

$\lambda = 0.135$ implies an average of roughly one jump every seven to eight years. Although not statistically significant, which is common given the limited number of jump observations, this magnitude is economically reasonable. Truly historic Atlantic hurricane seasons, such as 1933 and 2005, occur at a similar frequency in the historical record [7].

$\kappa = 0.202$ is the mean jump size and is positive, though not statistically significant. The positive sign is meaningful: jumps in the ACE index represent extraordinary bursts of hurricane activity, which push the index upward rather than downward. This is in contrast to indices where catastrophic events cause a drop, such as crop yield indices during droughts. The lack of statistical significance suggests that while jumps are on average positive, their exact size is highly unpredictable, which is consistent with the large estimated $\delta$.

To verify that MLE found the global maximum rather than a local one, we re-ran the optimization 10 times from different randomly picked starting values, generated by multiplying `param.init` by random draws from a uniform distribution on [0.5,1.5]. The log-likelihood values obtained across all 10 runs are reported below:

| Run | Log-Likelihood |
|---|---|
| 1 | $-208.9359$ |
| 2 | $-208.9359$ |
| 3 | $-208.9359$ |
| 4 | $-208.9359$ |
| 5 | $-208.9359$ |
| 6 | $-208.9359$ |
| 7 | $-208.9359$ |
| 8 | $-208.9359$ |
| 9 | $-208.9359$ |
| 10 | $-208.9359$ |

**Table 3:** Log-likelihood values across 10 random starting points

All 10 runs converge to exactly the same log-likelihood of -208.94, regardless of the starting point. This provides strong evidence that the optimisation found the global maximum rather than a local one.

# 4 Jump Risk as a Zero-Beta Event

A zero-beta event is an event whose risk is uncorrelated with the returns on the financial market, meaning it cannot be explained by systematic market movements and can therefore be diversified away in a well-constructed portfolio [10].

For jump risk in the ACE index to be a zero-beta event, the correlation between jump occurrences in the ACE index and financial market returns must be negligible in both directions.

Starting with the direction from financial markets to the ACE index: it is highly realistic to assume that financial markets have no influence on ACE jump risk. The ACE index is driven entirely by atmospheric physics and ocean temperatures, specifically sea surface temperatures, wind shear, and atmospheric pressure systems [6]. Interest rates, stock prices, investor sentiment, or monetary policy cannot alter these physical conditions. A financial crisis or a stock market crash has no bearing on whether an exceptional hurricane season occurs. This direction of the zero-beta assumption is therefore very realistic.

The other direction, whether an ACE jump event can move financial markets, requires more nuance. An exceptionally active hurricane season, such as 2005 with Hurricane Katrina, Rita and Wilma, can cause significant economic damage and temporarily affect the stock prices of insurers and reinsurers [3]. However, the impact on the broad financial market is limited for two reasons. First, hurricane damage is geographically concentrated in coastal regions of the US and the Caribbean, meaning the macroeconomic impact is local rather than systemic [12]. Second, the insurance and reinsurance industry exists precisely to absorb and redistribute these losses, preventing them from spilling over into the broader economy. Furthermore, CAT bond markets and reinsurance capital are globally diversified, meaning that even large insured losses are spread across international investors rather than concentrated in any single financial market [12]. As noted in Section 1, the SP500

ended 2005 with a positive annual return despite the most destructive hurricane season on record [3]. While one could argue that a sufficiently catastrophic season might produce second-order macroeconomic effects through supply chain disruptions or elevated government spending, any such effect would be geographically limited, leaving the long-run correlation with broad market returns small [16]. Any market impact from ACE jumps is therefore sector-specific rather than market-wide.

Overall, the zero-beta assumption for ACE jump risk is realistic. The physical nature of the index ensures that financial markets cannot influence hurricane activity, and the empirical evidence suggests that even extreme hurricane seasons do not produce lasting broad market shocks. This makes the ACE index a credible underlying for a CAT bond marketed as a zero-beta diversification instrument.

# 5 Suitable Market Index

A suitable market index to test whether the returns on the CAT index are uncorrelated with market returns is the S&P 500. While the MSCI World Index would, in principle, provide a more comprehensive representation of global financial markets, its limited historical availability restricts its usefulness for long-term empirical analysis. In contrast, the S&P 500 offers a substantially longer and more consistent time series, making it more suitable for this study.

The S&P 500 represents a broad and diversified portfolio of large-cap U.S. equities and is widely used as a proxy for overall market performance and systematic risk [17]. Although it is geographically concentrated in the United States, it still captures general market dynamics such as economic growth expectations, interest rate changes, and shifts in investor sentiment.

Using a broad market index ensures that the analysis reflects exposure to aggregate financial market movements rather than isolated or sector-specific shocks. At the same time, catastrophe risks such as tropical cyclones are geographically concentrated and driven by physical rather than economic factors [12]. Therefore, comparing CAT-related returns to a broad equity index helps avoid misleading correlations that may arise from regional overlap, allowing for a more accurate assessment of the independence between catastrophe risk and financial market performance.

To assess the relationship between the ACE index and financial markets, we compute the correlation between annual ACE log returns and annual S&P 500 log returns. Using returns rather than index levels ensures that the analysis compares changes in the two series on a consistent basis. The statistical testing of this correlation is discussed in the next section.

# 6 Correlation Test

To test whether the correlation between ACE and S&P 500 annual log returns is statistically different from zero, a Pearson correlation test is conducted. This test is appropriate because both variables are continuous and the objective is to assess the presence of a linear relationship. For the usual $t$-test approximation to be valid, the paired observations should also be approximately bivariate normal. Because both financial returns and catastrophe-related returns can display skewness

and heavy tails, this assumption is discussed critically in Section 7.

The hypotheses are defined as:

$$H_0 : \rho = 0$$
$$H_1 : \rho \neq 0$$

The test was implemented in R using the `cor.test()` function on the aligned annual log-return series of the ACE index and the S&P 500. The sample contains 73 paired annual observations covering 1950–2022. The function calculates the sample Pearson correlation coefficient and performs a two-sided test of the null hypothesis that the population correlation equals zero. The code is reported in Appendix B.

**Table 4:** Correlation between ACE and S&P 500 Annual Log Returns

| Statistic | Value |
|---|---|
| Correlation coefficient | 0.0514 |
| $t$-statistic | 0.4336 |
| Degrees of freedom | 71 |
| $p$-value | 0.6659 |
| 95% confidence interval | [-0.1808, 0.2782] |

At the 5% significance level, the $p$-value of 0.6659 exceeds 0.05. Therefore, the null hypothesis of zero correlation cannot be rejected. The estimated correlation is positive but very small, indicating that the sample provides no evidence of a meaningful linear relationship between annual ACE returns and annual S&P 500 returns.

# 7 Interpretation of the Statistical Test

The statistical test outcome is consistent with the argument made in Section 4 that ACE-related catastrophe risk can realistically be viewed as a zero-beta event, or at least as a risk with very low exposure to broad financial-market movements.

The estimated Pearson correlation between annual ACE log returns and S&P 500 log returns is 0.0514, which is close to zero. The correlation is not statistically significant at the 5% level, since the $p$-value equals 0.6659. In addition, the 95% confidence interval of [-0.1808, 0.2782] includes zero. The data therefore do not provide sufficient evidence to conclude that the true correlation differs from zero.

This result should still be interpreted carefully. Failing to reject the null hypothesis does not prove that the true correlation is exactly zero. It only means that the available sample does not provide sufficient statistical evidence of a non-zero linear relationship. Nevertheless, the point estimate itself is also very small, which strengthens the interpretation that ACE-related catastrophe risk has little exposure to broad equity-market movements.

This finding is consistent with the economic reasoning developed in Section 4. Financial markets cannot directly influence sea surface temperatures, wind shear or atmospheric pressure systems, while extreme hurricane seasons may still have limited indirect effects on insurers, reinsurers and affected sectors. Such effects are more likely to be sector-specific than market-wide.

The reliability of the Pearson correlation test also depends on its distributional assumptions. In practice, annual S&P 500 log returns may deviate from normality due to fat tails and occasional extreme observations [18], while ACE returns can also be strongly affected by extreme hurricane seasons. If normality is violated, the standard Pearson test may be less reliable [19]. Therefore, a Spearman rank correlation was also calculated as a robustness check. The Spearman correlation equals 0.0867 with a $p$-value of 0.4660. This result is also small and statistically insignificant, supporting the conclusion of a weak relationship between the two return series.

Overall, the Pearson and Spearman results point in the same direction. Together with the economic reasoning, they support the view that ACE-linked CAT bonds may offer diversification benefits and can reasonably be regarded as near-zero-beta instruments in practice, while not constituting definitive proof of complete independence.

# 8 CAT Bond Specification

Using the PJD parameters estimated in Section 3, we now specify a CAT bond written on the North Atlantic Accumulated Cyclone Energy (ACE) index. All remaining parameters needed for pricing are motivated below and summarised in Table 5.

**Table 5:** CAT bond parameters

| **Parameter** | **Value** | **Motivation** |
|---|---|---|
| Face value ($F$) | \$1,000 | Round denomination that does not affect relative pricing results. |
| Starting index value ($I_0$) | 100 | Consistent with ACE levels observed in recent seasons, as visible in Figure 1. |
| Trigger threshold ($K$) | 200 | Fewer than nine seasons since 1851 have exceeded this level (e.g. 1933 and 2005 [3, 7]), making it a genuine tail-risk threshold. Gives $I_0/K = 0.50 < 1$. |
| Maturity ($T$) | 1 year | The ACE index accumulates over one hurricane season, so a one-year horizon maps naturally to the risk period. |
| Loss fraction ($w$) | 1.0 | Full principal-at-risk: the entire face value is transferred to the issuer if the trigger is breached. |
| Risk-free rate ($r$) | 3% | As specified in the assignment instructions. |

The face value is set to $F = \$1{,}000$. This is a round denomination. The choice of face value

does not affect relative pricing results, since the bond price and yield spread are both expressed as functions of $F$ and scale proportionally with it.

The starting value is set to $I_0 = 100$. As visible in Figure 1, the ACE index has fluctuated considerably from year to year, but recent seasons have generally produced values in the range of roughly 50 to 175. A value of 100 therefore represents a moderately active season and is a reasonable starting point for the simulation. In practice, an issuer would set $I_0$ equal to the ACE value at the start of the coverage period.

The trigger is set at $K = 200$. As shown in Figure 1, fewer than nine Atlantic hurricane seasons since 1851 have produced ACE values above this level, most notably 1933 and 2005 [3, 7]. This makes $K = 200$ a genuine tail-risk threshold: it is not a level that a moderately active season would reach, but one that requires a historically exceptional combination of storm frequency and intensity. The resulting ratio $I_0/K = 100/200 = 0.50$ satisfies the required condition $I_0/K < 1$, and means the ACE index must double from its starting value before the bond is triggered.

The maturity is set to $T = 1$ year. The ACE index is recorded once per year, covering one full hurricane season. A one-year maturity therefore fits naturally with how the data is structured. Using a longer maturity would require modelling activity across multiple seasons; this goes beyond our scope. Furthermore, a longer maturity increases the probability that the threshold is breached at some point during the bond's lifetime, since there are more seasons over which an extreme hurricane year could occur. This additional risk would require a higher coupon rate to attract investors.

The loss fraction is set to $w = 1.0$. This means the bond is a full principal-at-risk instrument: if the ACE index exceeds $K = 200$ at any point during the hurricane season, investors lose their entire investment of \$1,000, and this capital is transferred to the issuer to cover catastrophe losses. We chose $w = 1.0$ rather than a partial loss structure because it maximises the capital available to the issuer in a disaster scenario. From the investor perspective, the risk of a complete loss of principal is compensated by a higher coupon, which will be reflected in the yield spread computed in Section 9.

The complete parameter set is: $F = \$1{,}000$, $I_0 = 100$, $K = 200$, $T = 1$, $w = 1.0$, $r = 3\%$. Combined with the PJD parameters from Section 3 ($\mu = 0.196$, $\sigma = 0.660$, $\delta = 1.550$, $\lambda_p = 0.135$, $\kappa = 0.202$), these fully determine the pricing problem. The only remaining input is the market price of the chosen CAT risk, $\lambda$, which is specified and discussed in Section 9.

# 9 CAT Bond Pricing Using the Vaugirard Simulation Approach

Using the Vaugirard simulation approach, we price the CAT bond specified in Section 8 for three different values of the market price of risk $\lambda$. All other parameters are as specified in Table 6. Before running the pricing, we performed a sensitivity analysis on the number of discretization steps $N$ and the number of simulation runs nsim. As noted in the comments of the code, these values are chosen ad hoc and require a sensitivity check. This analysis was done for $\lambda = 0.05$ and the results are reported in Table 6.

As visible in Table 6, the price estimates are stable across all combinations of $N$ and nsim, ranging

between \$677 and \$686. The choice of $N$ has little influence on the results, as prices remain similar when increasing $N$ from 100 to 1000. The number of simulation runs nsim has a clearer effect: a higher nsim consistently produces narrower confidence intervals. Based on these results, we select $N = 500$ and nsim $= 6{,}500$, as this combination produces one of the narrowest confidence intervals (CI width of 9.79) while keeping computation time manageable.

**Table 6:** Sensitivity analysis for $N$ and nsim ($\lambda = 0.05$)

| $N$ | nsim | Price (\$) | CI Lower | CI Upper | CI Width |
|---|---|---|---|---|---|
| 100 | 3500 | 683.83 | 677.62 | 690.04 | 12.41 |
| 100 | 5000 | 680.79 | 675.27 | 686.31 | 11.04 |
| 100 | 6500 | 682.29 | 677.34 | 687.23 | 9.89 |
| 500 | 3500 | 678.18 | 671.16 | 685.20 | 14.04 |
| 500 | 5000 | 681.23 | 675.66 | 686.81 | 11.14 |
| 500 | 6500 | 680.61 | 675.71 | 685.51 | 9.79 |
| 1000 | 3500 | 681.35 | 674.68 | 688.02 | 13.34 |
| 1000 | 5000 | 685.80 | 680.16 | 691.45 | 11.29 |
| 1000 | 6500 | 677.58 | 672.70 | 682.47 | 9.77 |

The market price of risk $\lambda$ is the parameter used in the Vaugirard risk-adjustment specification to translate the physical dynamics of the catastrophe index into the pricing measure. In the supplied implementation, $\lambda$ enters the drift adjustment through the term $\mu - \lambda\sigma$. Consequently, a larger value of $\lambda$ lowers the risk-adjusted drift of the ACE process, reduces the simulated trigger probability, and therefore raises the CAT bond price. This convention is important when interpreting the pricing results.

We consider three values of $\lambda$ to examine the sensitivity of the price to the assumed market price of catastrophe risk:

- $\lambda = 0.05$ (low): represents a relatively small risk adjustment.
- $\lambda = 0.30$ (moderate): represents an intermediate risk adjustment.
- $\lambda = 0.70$ (high): represents a comparatively strong risk adjustment and therefore provides a useful upper sensitivity case.

Table 7 reports the pricing results for each value of $\lambda$. For each run, we report the price and yield based on the PJD simulation with variance reduction technique (VRT), which is the most accurate estimate. The PJD prices with and without VRT are close to each other across all three runs, and the GBM simulation prices are close to the GBM formula prices, which confirms the simulation is working correctly.

Two patterns are immediately visible from Table 7. First, as $\lambda$ increases, the price of the CAT bond increases. Second, as $\lambda$ increases, the yield decreases. This is consistent with the inverse

**Table 7:** CAT bond pricing results for three values of $\lambda$

| | **PJD with VRT** | | **PJD without VRT** | |
|---|---|---|---|---|
| $\lambda$ | Price ($) | Yield | Price ($) | Yield |
| 0.05 (low) | 680.61 | 0.3548 | 695.59 | 0.3330 |
| 0.30 (moderate) | 743.21 | 0.2668 | 757.54 | 0.2477 |
| 0.70 (high) | 821.30 | 0.1669 | 829.06 | 0.1575 |

relationship between price and yield: as the price rises, the yield falls.

All three prices are below the face value of $1,000, which reflects the real possibility that the bond is triggered and investors lose their principal. The yield spread above the risk-free rate of 3% compensates investors for bearing this risk. For $\lambda = 0.05$ the yield is 0.3548, meaning investors earn approximately 35% above the risk-free rate. For $\lambda = 0.70$ this drops to 0.1669, or approximately 17% above the risk-free rate.

Among the three sensitivity cases, $\lambda = 0.05$ can be viewed as the most conservative choice for this analysis because it applies the smallest adjustment to the estimated physical dynamics of the ACE index. This is consistent with the economic argument that catastrophe risk is largely independent of broad financial-market movements. However, the precise calibration of $\lambda$ would in practice require market data on comparable CAT bonds and observed catastrophe-risk premia. The three values should therefore primarily be interpreted as sensitivity scenarios rather than directly observable market estimates.

# Use of AI Statement

Claude and ChatGPT were used to assist in structuring the argument and identifying relevant data points from public sources. All analytical conclusions, interpretation and final text are written by myself.

# A PJD Parameter Estimation

```
library(maxLik)
source( functions.R )
source( LL.R )

# Load the ACE data and compute log returns
cat.data = read.csv( ace-north-atlantic-hurricanes.csv )
ACE = cat.data$Accumulated.cyclone.energy.index
return.cat = log(ACE[2:length(ACE)] / ACE[1:(length(ACE)-1)])
N = length(return.cat)
delta.t = 1 # annual data

summary(return.cat)

# Try different epsilon values to find a good split between jumps and
   diffusion
for (eps in c(0.8, 1.0, 1.2)) {
  jumpindex_temp = which(abs(return.cat) > eps)
  cat( Epsilon: , eps,
       |␣Jumps: , length(jumpindex_temp),
       |␣Lambda: , length(jumpindex_temp) / (N * delta.t),  \n )
}

# We go with epsilon = 1.0 as it gives a reasonable number of jumps
epsilon = 1.0
jumpindex = which(abs(return.cat) > epsilon)
diffusionindex = which(abs(return.cat) <= epsilon)
Rjumps = return.cat[jumpindex]
Rdiffusion = return.cat[diffusionindex]

# Compute starting values for MLE
sigma_hat = sd(Rdiffusion) / sqrt(delta.t)
mu_hat = (2*mean(Rdiffusion) + (sigma_hat^2)*delta.t) / (2*delta.t)
lambdap_hat = length(jumpindex) / (N * delta.t)
delta_hat = sqrt(abs(var(Rjumps) - sigma_hat^2*delta.t))
kv_hat = mean(Rjumps) - (mu_hat - sigma_hat^2/2)*delta.t

param.init = c(mu_hat, sigma_hat, delta_hat, lambdap_hat, kv_hat)
print(param.init)

# Run MLE
outp = maxLik(LL, start=param.init, method= BHHH , control=list(iterlim
   =1000))
summary(outp)
est.param = outp$estimate

```

```
# Verify we found the global maximum by trying 10 different starting points
set.seed(123)
results = list()
for (i in 1:10) {
  start_perturbed = param.init * runif(5, 0.5, 1.5)
  try({
    fit = maxLik(LL, start=start_perturbed, method= BHHH , control=list(iterlim=1000))
    results[[i]] = c(fit$maximum, fit$estimate)
  })
}

for (i in 1:10) {
  if (!is.null(results[[i]])) {
    cat( Run , i,  -␣LogLik: , results[[i]][1],  \n )
  }
}
```

Listing 1: MLE estimation of PJD parameters used in Section 3

# B Correlation Analysis

```
library(quantmod)

# S&P 500 log returns
getSymbols( ^GSPC , src =  yahoo , from =  1950-01-01 )
prices <- Ad(GSPC)
annual_returns <- periodReturn(prices, period =  yearly , type =  log )
years <- format(index(annual_returns),  %Y )
sp_returns <- data.frame(
  Year = as.numeric(years),
  SP500_log_return = as.numeric(coredata(annual_returns))
)

# ACE data
ace_data <- read.csv( ace-north-atlantic-hurricanes.csv )
ace_na <- ace_data[, c( Year ,  Accumulated.cyclone.energy.index )]
colnames(ace_na) <- c( Year ,  ACE )

# Compute annual ACE log returns
ace_na <- ace_na[order(ace_na$Year), ]
ace_na$ACE_log_return <- c(
  NA,
  log(ace_na$ACE[-1] / ace_na$ACE[-nrow(ace_na)])
)

```

```r
# Align the two annual return series
df <- merge(
  ace_na[, c( Year ,  ACE_log_return )],
  sp_returns,
  by =  Year
)
df <- na.omit(df)

# Pearson correlation test
cor.test(df$ACE_log_return, df$SP500_log_return, method =  pearson )

# Optional robustness check
cor.test(df$ACE_log_return, df$SP500_log_return,
         method =  spearman , exact = FALSE)
```

Listing 2: Correlation analysis used in Section 6

# C CAT Bond Pricing Code

The code below is from `part4_cat_price_vrt.R`. The following changes were to the original code:

- `doMC` was replaced by `doParallel`, since `doMC` is not available on Windows.
- The MLE parameters are coded directly instead of reading from `est_param.txt`.
- The functions `mut()`, `sigmat()` and `lambdat()` are defined directly in the script instead of loading `functions.R`.
- The CAT bond parameters are set to our own values as specified in Section 8.
- A sensitivity analysis loops over $N$ and nsim was added, as recommended by the comments in the original code.

```r
# Load libraries
library(foreach)
library(doParallel)
registerDoParallel(cores=4)

# MLE parameters from Section 3 (Table 2)
mu     = 0.196
sigma  = 0.660
delta  = 1.550
lambda.p = 0.135
kv     = 0.202

# CAT bond parameters from Section 8 (Table 5)
F  = 1000   # face value
I0 = 100    # starting value of the ACE index
```

```r
K  = 200    # trigger threshold (I0/K = 0.50 < 1)
T  = 1      # maturity in years
w  = 1.0    # full principal-at-risk
r  = 0.03   # risk-free rate (as specified in assignment)

# Functions defined directly instead of source( functions.R )
mut     <- function(t){ mu }
sigmat  <- function(t){ sigma }
lambdat <- function(t){ lambda }

# Sensitivity analysis on N and nsim
# We test combinations of N in (100, 500, 1000) and nsim in (3500,
   5000, 6500) for a fixed lambda = 0.05.

lambda = 0.05
N_values    <- c(100, 500, 1000)
nsim_values <- c(3500, 5000, 6500)
results <- data.frame()

for(N in N_values){
  for(nsim in nsim_values){

    delta.t <- T/N

    # GBM formula (needed for variance reduction technique)
    delta1 <- mu - lambda*sigma
    d1 <- (log(I0/K) + (delta1 - sigma^2/2)*T) / (sigma*T^0.5)
    d2 <- (log(I0/K) - (delta1 - sigma^2/2)*T) / (sigma*T^0.5)
    E  <- pnorm(d1) + (I0/K)^(1 - 2*delta1/sigma^2)*pnorm(d2)
    EPV.formula <- exp(-r*T)*(F - w*F*E)

    risk.index1 <- matrix(0,(N+1),1)
    risk.index1[1] <- I0
    risk.index2 <- matrix(0,(N+1),1)
    risk.index2[1] <- I0

    sim.res <- foreach(p=1:nsim, .combine=rbind) %dopar% {

      set.seed(nsim*p)
      Z1 <- rnorm(N)
      N.Poisson <- rpois(N, lambda.p*delta.t)

      for(i in 1:N){
        if(N.Poisson[i]>0){
          Z2 <- exp(rnorm(N.Poisson[i])*delta + kv) - 1
        }else{
          Z2 <- 0
        }
```

```r
        incr1 <- (mut(delta.t*i) - lambdat(delta.t*i)*sigmat(delta.t*i)
           )*delta.t +
                sigmat(delta.t*i)*Z1[i]*sqrt(delta.t) + sum(Z2)
        risk.index1[i+1] <- risk.index1[i] + incr1*risk.index1[i]

        incr2 <- (mut(delta.t*i) - lambdat(delta.t*i)*sigmat(delta.t*i)
           )*delta.t +
                sigmat(delta.t*i)*Z1[i]*sqrt(delta.t)
        risk.index2[i+1] <- risk.index2[i] + incr2*risk.index2[i]
      }

      PV.jumps    <- exp(-r*T)*(F - w*F*(max(risk.index1)>=K))
      PV.brownian <- exp(-r*T)*(F - w*F*(max(risk.index2)>=K))
      c(PV.jumps, PV.brownian, EPV.formula)
    }

    # Variance reduction technique (as in part4_cat_price_vrt.R)
    diffprice.GBM  <- sim.res[,2] - sim.res[,3]
    coeff          <- cov(sim.res[,1], diffprice.GBM) / var(diffprice.
       GBM)
    adjustedprice.PJD <- sim.res[,1] - coeff*diffprice.GBM
    value.sim1     <- mean(adjustedprice.PJD)

    # Confidence interval
    se       <- sd(adjustedprice.PJD) / sqrt(nsim)
    ci_lower <- value.sim1 - 1.96*se
    ci_upper <- value.sim1 + 1.96*se
    ci_width <- ci_upper - ci_lower
    yield    <- -log(value.sim1/F)/T - r

    results <- rbind(results, data.frame(
      N = N, nsim = nsim,
      Price    = round(value.sim1, 2),
      CI_Lower = round(ci_lower, 2),
      CI_Upper = round(ci_upper, 2),
      CI_Width = round(ci_width, 2),
      Yield    = round(yield, 4)
    ))
  }
}
print(results)

# Pricing for three values of lambda, based on the sensitivity analysis
    we use N = 500 and nsim = 6500.
# The code below is run three times, changing lambda each time for
   lambda = 0.05 (low), lambda = 0.30 (moderate), lambda = 0.70 (high)

N    = 500
```

```r
nsim = 6500

# Change lambda for each run
lambda = 0.05  # change to 0.30 or 0.70 for the other runs

# GBM formula (needed for variance reduction technique)
delta1      <- mu - lambda*sigma
d1          <- (log(I0/K) + (delta1 - sigma^2/2)*T) / (sigma*T^0.5)
d2          <- (log(I0/K) - (delta1 - sigma^2/2)*T) / (sigma*T^0.5)
E           <- pnorm(d1) + (I0/K)^(1 - 2*delta1/sigma^2)*pnorm(d2)
EPV.formula <- exp(-r*T)*(F - w*F*E)

delta.t <- T/N
risk.index1 <- matrix(0,(N+1),1)
risk.index1[1] <- I0
risk.index2 <- matrix(0,(N+1),1)
risk.index2[1] <- I0

# PJD simulation with variance reduction
sim.res <- foreach(p=1:nsim, .combine=rbind) %dopar% {

  set.seed(nsim*p)
  Z1 <- rnorm(N)
  N.Poisson <- rpois(N, lambda.p*delta.t)

  for(i in 1:N){
    if(N.Poisson[i]>0){
      Z2 <- exp(rnorm(N.Poisson[i])*delta + kv) - 1
    }else{
      Z2 <- 0
    }
    incr1 <- (mut(delta.t*i) - lambdat(delta.t*i)*sigmat(delta.t*i))*
       delta.t +
              sigmat(delta.t*i)*Z1[i]*sqrt(delta.t) + sum(Z2)
    risk.index1[i+1] <- risk.index1[i] + incr1*risk.index1[i]

    incr2 <- (mut(delta.t*i) - lambdat(delta.t*i)*sigmat(delta.t*i))*
       delta.t +
              sigmat(delta.t*i)*Z1[i]*sqrt(delta.t)
    risk.index2[i+1] <- risk.index2[i] + incr2*risk.index2[i]
  }

  PV.jumps     <- exp(-r*T)*(F - w*F*(max(risk.index1)>=K))
  PV.brownian <- exp(-r*T)*(F - w*F*(max(risk.index2)>=K))
  c(PV.jumps, PV.brownian, EPV.formula)
}

# Variance reduction technique
```

```r
diffprice.GBM     <- sim.res[,2] - sim.res[,3]
coeff             <- cov(sim.res[,1], diffprice.GBM) / var(diffprice.
   GBM)
adjustedprice.PJD <- sim.res[,1] - coeff*diffprice.GBM
value.sim1        <- mean(adjustedprice.PJD)

# Prices 1 and 2 should be close, prices 3 and 4 should be close
cat( Lambda: , lambda,  \n )
cat( Price (PJD with VRT): , value.sim1,  \n )
cat( Price (PJD without VRT): , colMeans(sim.res)[1],  \n )
cat( Price (GBM simulation): , colMeans(sim.res)[2],  \n )
cat( Price (GBM formula): , colMeans(sim.res)[3],  \n )

# Yields
yield.sp1 <- -log(value.sim1/F)/T - r
yield.sp2 <- -log(colMeans(sim.res)[1]/F)/T - r
yield.sp3 <- -log(colMeans(sim.res)[2]/F)/T - r
yield.sp4 <- -log(colMeans(sim.res)[3]/F)/T - r

cat( Yield (PJD with VRT): , yield.sp1,  \n )
cat( Yield (PJD without VRT): , yield.sp2,  \n )
cat( Yield (GBM simulation): , yield.sp3,  \n )
cat( Yield (GBM formula): , yield.sp4,  \n )
```

Listing 3: Sensitivity analysis and CAT bond pricing used in Section 9

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